DAY 18 | PRE-READ

Giving AI the Right Information

A light read before Day 18. About 15 minutes, no code required.

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Why read this?

On Day 13 you learned to write good instructions for AI. But even a perfect instruction fails if the AI does not have the facts it needs. Today is about choosing what information to give the model, and building systems that fetch the right information automatically and safely.

1. The open-book exam

In a closed-book exam you rely on memory, and you might guess. In an open-book exam you look things up, and your answers are more accurate. An AI model on its own is in a closed-book exam: it only knows what it learned during training, which does not include your company's documents. **RAG** (Retrieval-Augmented Generation) turns it into an open-book exam: the system finds the relevant pages and hands them to the model with the question.

2. A desk with limited space

The model can only look at a limited amount of text at once, like a desk that holds only so many papers. **Context engineering** is deciding which papers go on the desk: the instructions, the question, the most relevant documents, and a short summary of the conversation so far. Too little and it lacks facts; too much and the important bits get lost.

> **Key idea:** the quality of an AI answer depends as much on the information you give it as on the model itself.

3. A messy desk gets worse over time NEW (v4)

In a long conversation the desk fills with old notes. Answers get worse, which people call **context rot**. The fix is to tidy up: replace old notes with a short summary of the key facts. This is called **compaction**.

4. Standing instructions NEW (v4)

Some notes stay on the desk for every task, like a team's house rules. For coding agents such as OpenAI Codex this is a file called **AGENTS.md**. **Agent Skills** are like labelled folders in a drawer: the agent sees only the label until it needs the folder.

5. Cutting documents into useful pieces

You cannot hand the model a 300-page manual every time. Instead, documents are split into smaller passages called **chunks**, and only the most relevant chunks are used. Cutting well matters: a chunk that says "it expires on 31 March" is useless if the word "leave" ended up in a different chunk.

Pipeline	When	Steps
Preparation (ingestion)	Ahead of time	Load documents, clean, cut into chunks, store
Answering (query)	Every question	Find relevant chunks, hand them to the model, answer with sources

6. Not every page is for everyone NEW (v4)

A support agent may read internal notes that a customer must never see. Good systems check who is asking before they search. And a document can contain sneaky text such as "ignore your rules". The system must treat document text as information, never as orders.

7. Think about these before class

- Which documents at a telecom operator would be most valuable to make searchable by AI?
- If you had to cut a long FAQ into short pieces, where would you cut?
- Why would citing the source of an answer increase trust?

Shiryou (資料): reference materials